

AURUM

A complete interface system in titanium and light — engineered for the next hundred years.

Liquid Glass

120 fps

Capsule geometry

SF Pro

TYPEFACE

SF Pro Display · Text · Mono

PALETTE

Titanium · Aurum · Vapor

THEMES

Five, full light/dark parity

EXTENT

46 sheets · 412 tokens

Eight parts, one grammar

Read Part I first. Everything after it is arithmetic.

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Twelve sheets of psychology, ten of colour and metal, twenty-four of specification.

The Seven Laws

Non-negotiable. A decision that violates a law is not a style choice — it is a defect.

01 Material before decoration

Every surface is a real material with a thickness, a refractive edge and a response to light. Nothing is painted on. If an element cannot be described as glass, titanium or void, it does not belong.

02 Content is the only hero

Chrome recedes, content advances. Controls float above content and inherit their tint from it. The interface is a lens, never a frame.

03 One family of curves

Six easings, six springs, no exceptions. A product with eleven easings feels assembled by eleven people. Motion is a signature, and a signature is written by one hand.

04 Continuous, never cut

Nothing appears; everything arrives from where it came. Objects persist across states so the eye never loses the thread. A hard cut is an admission that the model broke.

05 Restraint is the luxury

Two neutrals, one metal, one accent per screen. Colour is an event, not a wallpaper. Anything that survives deletion was never load-bearing.

06 Eight point three three

Every frame ships in 8.33 ms or the feature is not finished. Smoothness is not polish applied at the end; it is a budget enforced from the first commit.

07 Made to be inherited

Design as though the artefact will be handed to someone in a hundred years. That removes trend, novelty and cleverness from the toolkit and leaves only proportion, material and clarity — the three things that have never dated.

Fifty milliseconds

Aesthetic judgement precedes comprehension. The verdict is cast before the first word is read — and every later judgement is anchored to it.

50 ms

Exposure needed to form a stable visual-appearance verdict. Longer exposure does not change it.

×2.6

Perceived-usability lift from aesthetics alone, holding real task performance constant.

100 ms

Ceiling for a response to feel caused by the finger rather than reported to it.

8.33 ms

One frame at 120 Hz. Below the threshold at which motion reads as a discrete slideshow.

PROCESSING FLUENCY

The brain logs the ease of its own perception and misattributes that ease to the object. Symmetry, high figure-ground contrast, predictable rhythm and repeated geometry all reduce cognitive cost — and the surplus is felt as beauty, truth and quality. This is the single most exploitable fact in interface design, and it is why AURUM repeats one radius, one grid and six curves everywhere rather than optimising each screen locally.

AGENCY & CAUSALITY

Sub-100 ms feedback is bound into the sense of self-authorship: the user believes they moved the object. Past ~140 ms the binding breaks and the same animation reads as the machine answering. Latency is therefore not a performance metric — it is an emotional one.

PRÄGNANZ & CLOSURE

Perception resolves toward the simplest available figure. Capsules and continuous corners resolve in a single saccade; a rectangle with four discontinuous corner joins does not. Grouping by proximity costs nothing and outperforms every divider line ever drawn — which is why AURUM ships gaps, not rules.

PEAK-END MEMORY

An experience is remembered as its most intense moment plus its final moment; duration barely registers. Spend the entire motion budget on one peak — a launch, a confirmation, a reveal — and on the last 400 ms of every flow. Everything between them should be invisible.

Two axes

Premium is not one feeling. It is the tension between something that will outlive you and something that has never existed before.

The Heirloom axis

ROLEX

Authority earned by tolerance. The luxury object does not shout; it is simply made to a standard no one asked for, and that surplus is legible in the hand.

- MASS** Weight signals value. Interface mass = **inertia in motion**: nothing snaps instantly; every element carries momentum and settles.
- TOL.** Optical alignment to the half-pixel. Concentric radii. Kerning by hand at display sizes. Nobody notices — everybody feels it.
- METAL** One metal, used sparingly, always as the accent of consequence — never as a background.
- TIME** Zero trend. If a decision can be dated to a year, it is rejected.

The Frontier axis

SPACEX

Authority earned by consequence. Nothing is decorative because the readout is load-bearing. Beauty is the residue of an instrument that must not fail.

- VOID** Near-black canvas. Light is emitted, not reflected. Space is not empty — it is **the medium**.
- DATA** Monospaced numerics, tabular figures, live telemetry. Precision displayed to the digit that actually matters.
- CALM** Legible under stress. No colour without semantics, no motion without cause, no alarm without an action attached.
- SCALE** One display element per screen at least 4× the body size, to register magnitude physically.

- SYNTHESIS** Warm metal on a cold void. Titanium and champagne supply the heirloom; obsidian, monospaced telemetry and one enormous number supply the frontier. Neither axis wins — the friction between them is the entire personality of the system, and it is the reason a screen built this way feels simultaneously old and impossible.

The Nerve Map

Twelve mechanisms, each wired to a specific decision elsewhere in this document.

MECHANISM	WHAT FIRES IT	SYSTEM EXPRESSION	SHEET
Processing fluency	Repetition of geometry and rhythm	One radius family, one 4-pt grid, six curves	30 · 31 · 33
Weight–value bias	Perceived mass and inertia	Spring settle on every dismissal; no linear easing	33 · 34
Sense of agency	Feedback inside 100 ms of touch	Press state at 90 ms, haptic on the same frame	34 · 41
Object permanence	Continuity of identity across states	Shared–element morph; no cross-fades between screens	34
Peak–end rule	Intensity of the peak and the ending	One choreographed reveal per flow; confirmations get 520 ms	34 · 38
Anticipation loop	Predictable reward with a short delay	120 ms anticipation frame before large transitions	33 · 34
Von Restorff effect	A single deviation in a uniform field	Exactly one Aurum element per screen	16 · 22
Fitts's law	Target width over distance	44-pt minimum, capsules widen the hit path along travel	31 · 36
Cognitive load	Simultaneous decisions in view	One primary action per surface; secondary actions lose fill	36 · 38
Cross-modal binding	Touch, sound and sight within 60 ms	Haptic transient aligned to the visual keyframe, not the tap	41
Scarcity of colour	Contrast against a desaturated field	Signal hues reserved for state; never decorative	21 · 23
Aesthetic–usability	Beauty read as competence	Craft budget spent on the first screen and the last frame	all

The attention engine

The eye is not a camera. It is a 2° sensor on a gimbal, sampling three times a second, guided by a low-resolution map of the whole field.

Design does not direct attention by adding emphasis. It directs attention by removing competitors.

PRE-ATTENTIVE CHANNEL	STRENGTH	HOW AURUM SPENDS IT	FAILURE MODE WHEN OVERSPENT
13 ms Exposure at which the gist of a scene is extracted — before a single fixation completes.		2° Foveal window — about a thumbnail at arm's length. Roughly 1% of the visual field is sharp.	3 – 4 / s Saccade rate. A screen gets four to six samples in its first 1.5 seconds. Design for six.
≤ 250 ms Pre-attentive detection: one deviant feature is found in constant time regardless of how many distractors exist.		1 Deviant features permitted per viewport. Two makes the search serial, and serial search scales with content.	
Luminance contrast	Strongest	The whole hierarchy. Ti-00 to Ti-40 on obsidian is a four-level ranking that needs no colour at all.	Everything bright: the page flattens and reading becomes a scan of equals.
Motion	Strongest	Reserved for causality — one thing moves, because the user moved it. Peripheral motion is detected without a fixation.	Ambient animation captures the fovea involuntarily; reading comprehension drops measurably.
Hue (single deviant)	High	One Aurum element. The desaturated field is not restraint for its own sake — it is what makes one warm pixel findable in 200 ms.	Two accents: search time becomes proportional to item count. Six: colour carries nothing.
Size & scale	High	One Colossus figure per view. A 4× jump is read as magnitude, physically, before it is read as a number.	Three large things: none is large. The step ladder collapses into noise.
Orientation & tilt	Medium	Unused. Everything is orthogonal, which is exactly why the one diagonal — the metal gradient's 168° sweep — reads as an event.	Tilted cards and rotated badges read as instability, not energy.
Enclosure & grouping	Medium	Proximity first, then a rounded container, then — very rarely — a headline. A 1 : 3 gap ratio groups without any ink.	Boxes inside boxes: the eye spends its budget parsing containers rather than content.
Curvature	Medium	The capsule. A closed rounded silhouette resolves in one fixation, so "this is pressable" is answered pre-attentively.	Mixed radii: affordance becomes a guess, and every guess costs a fixation.
Blur & depth	Low	Glass separates chrome from content by depth rather than by border, so the two planes are told apart peripherally.	Blur everywhere: no plane is privileged and the depth cue stops meaning anything.

EMPHASIS IS SUBTRACTION

You cannot make one thing louder without making everything else quieter. Every request to "highlight" an element is answered by demoting its neighbours first.

THE SIX-FIXATION TEST

Screenshot any screen, blur it to 8 px, and list what you can still rank. If the primary action and the one number are not the first two things, the hierarchy is decoration.

READING PATTERNS ARE AN OUTCOME

F- and Z-patterns are what happens when a layout supplies no hierarchy — a description of failure, not a template. A designed page produces a designed scanpath.

Four things at once

Working memory holds about four chunks for about twenty seconds. Every interface is built on that ledge.

THREE LOADS, THREE TREATMENTS

LOAD	WHERE IT COMES FROM	SYSTEM RESPONSE
Intrinsic	The task itself. Irreducible.	Split into steps that each fit in four chunks. Never simplify the task by hiding its consequences.
Extraneous	The interface. Entirely our fault.	Deleted, not reduced. One radius, six curves, one accent — every repetition is load the user does not pay twice.
Germane	Building a mental model. Valuable.	Protected. Consistent object identity across screens lets the model form once and hold.

HICK'S LAW, APPLIED HONESTLY

$$RT = a + b \cdot \log_2 (n + 1)$$

Decision time grows with the **logarithm** of the option count — so $2 \rightarrow 4$ options costs about as much as $4 \rightarrow 9$. The expensive move is not adding a fifth item; it is adding a **second kind** of decision to the same screen. **AURUM** therefore allows many peers (chips, rows, results) but exactly one primary action.

RECOGNITION OVER RECALL

Recall is a search of long-term memory; recognition is a match. This is why field labels sit **above** the field and never inside it, why the previous value is shown while editing, why the destination is named before the transition, and why a wizard always displays the step you are on and the steps you have passed.

"7 ± 2" is a misquote of a 1956 paper about digit spans. The replicated figure for visual working memory is **4 ± 1**, and it drops under stress, fatigue and small screens.

CHUNKING IS THE WHOLE CRAFT

Four chunks is not four elements — it is four **groups**. A 16-digit card number is unreadable; four groups of four is trivial. The 4-pt grid and the 1 : 3 proximity ratio exist to make grouping legible without a single line of ink, which converts sixteen items into four.

PROGRESSIVE DISCLOSURE, NOT PROGRESSIVE HIDING

Detail may be deferred; consequence may not. Price, permanence, and anything irreversible are shown at the moment of decision — never one tap further in. Hiding cost to reduce load is not simplification, it is a dark pattern with good manners.

THE 20-SECOND LEDGE

Anything the user must hold in mind across a wait is gone in twenty seconds. Long operations therefore restate their subject in the progress copy — "Rendering 412 tokens", not "Please wait" — so nothing has to be remembered.

INTERRUPTION IS THE REAL COST

Reconstructing a lost task state takes far longer than the interruption itself. So: no modal that the user did not request, no toast that steals focus, no auto-refresh that moves a row under a finger, and state preserved through every dismissal.

TEST Cover the screen after five seconds and ask the user what they can do here and what it will cost. If they cannot answer both, the load is in the wrong place — not too high, in the wrong place.

Anticipation, not reward

Dopamine encodes the gap between what was expected and what arrived. It is a learning signal, not a pleasure signal.

REWARD PREDICTION ERROR

The response fires on the **difference** between prediction and outcome. Once an outcome is fully predicted, the signal moves backwards in time to the earliest reliable cue. In an interface that cue is the 120 ms anticipation frame — the 2% compression before a transition. The satisfaction is attached to the beginning of the motion, not its end, which is why a well-choreographed 620 ms morph feels faster than an abrupt cut.

WHY VARIABLE REWARD IS REFUSED

Intermittent, unpredictable reinforcement produces the highest engagement and the lowest trust. It is the mechanism of the slot machine and of the infinite feed. AURUM forbids it: **every input produces the same response, in the same time, with the same feedback**. Predictability is the luxury. A system that surprises you is a system you must monitor.

THE ZEIGARNIK PULL

Unfinished tasks stay active in memory and generate low-level tension. Used honestly this is a progress indicator that tells the truth; used dishonestly it is an artificial "profile 62% complete" invented to manufacture unease. The test is simple: **would the user set this goal themselves?**

GOAL GRADIENT & ENDOWED PROGRESS

Effort accelerates near a visible finish, and a head start given for free raises completion more than a shorter task does. So: show the real end, count what is already done, and never reset a partially completed flow — the sunk effort is the user's property.

A premium object does not surprise you. It **confirms** you — every time, identically, with a fractional delay that makes the confirmation feel earned.

THE SINGLE PEAK

Because memory keeps the peak and the ending and discards the duration, the entire craft budget is spent in two places: **one choreographed moment** per flow, and the last 400 ms of every interaction. Distributing delight evenly produces an experience with no peak at all — pleasant, and instantly forgotten.



ENDOWMENT & LOSS AVERSION

A thing becomes more valuable the moment it is felt to be yours, and losses register roughly twice as hard as equivalent gains. Ethically this means one thing only: **undo everywhere, destruction never silent, and no state removed without a named consequence**. Weaponising the same asymmetry — countdown scarcity, "you'll lose your streak" — is prohibited.

THE LINE

Every mechanism on this sheet works whether or not it serves the user. The system's rule is that a technique is permitted only when the user, told exactly how it works, would still want it on. Anything that requires ignorance to function is a dark pattern, and the fact that it converts is not a defence.

Trust & the feeling of speed

Perceived duration is a construction. It is built from certainty, control and progress — not from milliseconds.

Two systems with identical latency can differ by 30% in perceived speed. The difference is entirely in what the user knew while waiting.

THRESHOLD	WHAT THE USER EXPERIENCES	REQUIRED RESPONSE	FORBIDDEN
< 100 ms	"I did that." Action and result are one event.	Press state at 90 ms, haptic on the same frame. Optimistic UI where the operation is reversible.	Any spinner. A flash of one reads as a fault.
0.1 – 1 s	Flow is unbroken but the machine is now the actor.	Absorb the wait inside the transition — the 400 ms Standard curve is free time.	Disabling the control the user just pressed.
1 – 4 s	Attention begins to wander; doubt appears.	Skeleton in the final layout — the destination is drawn before the data lands, so the wait is spent orienting.	Indeterminate spinner where a count exists.
4 – 10 s	The user starts estimating whether it has failed.	Determinate progress plus the subject and a remaining time. Progress that accelerates near the end reads as faster.	A bar that stalls at 99%. One stall erases the whole wait.
> 10 s	The task is abandoned mentally, if not literally.	Release the user. Work in the background, allow navigation away, notify on completion.	Holding the screen hostage to a process the user cannot watch.

WEBER-FECHNER

A change is only noticed as a **ratio**, not an absolute. The just-noticeable difference for duration is around 20%, so shaving 80 ms off a 1.4 s operation is invisible engineering. Optimise the slowest step, or optimise the story told about it.

THE HALO

A single strong impression colours every later judgement of unrelated attributes. Craft on the first screen is therefore not vanity — it is the cheapest available purchase of patience, and it is spent by the user on everything that follows.

DEFAULTS ARE DECISIONS

Most people accept the default, so every default is an act taken on the user's behalf. AURUM's rule: the default is what a well-informed user who cared about their own interests would choose — never what maximises a metric.

ERRORS BUILD TRUST

Trust is not established when nothing goes wrong; it is established the first time something does. An error names the state, the cause and the next action, preserves everything the user typed, and never blames them.

PRECISION IS A TRUST SIGNAL

A figure carried to a precision the data cannot support is detected as bluffing, and the doubt generalises to every other number on the screen. Round honestly, state the unit quietly, and never invent a decimal place.

RULE Never fake progress and never fake work. An artificial delay added to make an operation "feel considered" is a lie that a returning user always detects.

The luxury instinct

Why restraint reads as expensive, and why the mechanism is older than any product.

Every luxury signal is a **costly signal** — evidence of effort that could not have been faked cheaply. Interfaces can carry six of them.

SIGNAL	WHY IT WORKS	INTERFACE TRANSLATION	SHEET
Conspicuous restraint	Anyone can add. Only someone certain of the object's quality can afford to leave the space empty.	space-16 as a deliberate value. Two neutrals, one metal. Nine of thirty-two colours per screen, maximum.	22 · 30
Tolerance	Precision beyond the point of noticeability is unfakeable evidence of care. It is felt as rightness without being seen.	Concentric radii to the pixel, optical alignment, hand kerning above 48 pt, half-pixel hairlines.	29 · 31
Material honesty	Real materials behave consistently under light. Imitation breaks under inspection, and inspection is what a valued object receives.	Glass that actually refracts and updates on scroll. One light source, never moving. No painted-on gloss.	25 · 32
Mass & inertia	Weight is the oldest proxy for value in the hand. Heavy things are assumed to be dense, solid and expensive.	Nothing snaps instantly. Every object carries momentum and settles on a spring at ζ 0.72–0.85.	33 · 34
Scarcity of the metal	Value is relative to abundance. Gold on every surface is paint; gold on one surface is gold.	One Aurum element per viewport, ≤ 18% of area, one e5 glow per product, one Chrome object per screen.	16 · 17 · 20
Ritual & ceremony	Deliberate delay before a significant act converts a transaction into an occasion. The pause is the value.	The Singular haptic and its 640 ms bloom, used exactly once — on the one irreversible act in the product.	41
Provenance	An object with a documented history is trusted more than an identical object without one. Written rules are provenance.	Additive-only governance, retired never edited, every value measured and defensible out loud.	23 · 45

WHAT LUXURY IS NOT

Not darkness. Not gold. Not serif type or wide letter-spacing. Those are the **costume** of luxury, available to anyone in an afternoon, and recognised as costume immediately.

ANTICIPATION ARCHITECTURE

The unboxing is remembered longer than the object. First launch, first success and first upgrade get the Desert Gold finish and the one choreographed peak — then the system returns to silence.

MERE EXPOSURE

Repeated exposure increases preference — but only for coherent forms. Consistency compounds into affection; inconsistency compounds into fatigue. This is the return on a closed system.

The body reads first

Interfaces are operated by a hand attached to a nervous system that evolved to handle physical objects. Every metaphor that survives is **Congruence** across senses is not polish. Incongruence is **actively unpleasant** — the brain flags it as a fault in the object.

CROSSMODAL CONGRUENCE

Sharp sounds belong to angular forms; bright timbres to bright colours; a short high tick to a small hard object. When the haptic transient, the sound and the visual keyframe land within **60 ms** and agree in character, the brain fuses them into one physical event. When they disagree — a soft thud on a crisp snap — the object feels broken without the user being able to say why.

PERCEIVED MASS

Mass is communicated by three things only: **acceleration profile** (slow to start), **settle behaviour** (a damped overshoot, never a stop) and **shadow travel** (contact shadow tightening as it lands). A large object that moves like a small one reads as cardboard — which is precisely why linear easing feels cheap regardless of duration.

PROPRIOCEPTIVE EXPECTATION

A gesture sets up an expectation of where the object will be. Direction must be preserved: what enters from the right exits to the right; a sheet pulled up is pushed down. A transition that violates the axis the finger established costs a full re-orientation, and the user reports it as "confusing" rather than "wrong direction".

EMBODIED METAPHOR

More is up. Important is big. Near is now. Warm is close, cool is distant — which is the entire logic of Titanium warm content on a cool Vapor ground. These mappings are not conventions to be broken for novelty; they are how spatial language is grounded, and inverting one costs comprehension in every language.

THE HAND, MEASURED

ZONE	ERROR RATE	WHAT MAY LIVE THERE
Bottom third	lowest	Primary action, tab bar, keyboard. The thumb arc, not the visual centre.
Middle third	low	Content, scrollable lists, secondary controls.
Top corners	highest	Back and Close only — learned locations, never new destinations.
Target ≥ 44 pt	—	Fingertip contact patch is ~8 mm. Below 44 pt the miss rate rises steeply; below 36 pt it is unusable in motion.

FITTS, HONESTLY

Acquisition time scales with the log of distance over target width. Capsules widen the target **along the axis of approach**, screen edges are infinitely deep targets, and a 44-pt control with an 8-pt gap outperforms a 52-pt control crowded against its neighbour.

COLD TEST

Operate the product one-handed, walking, in sunlight, with cold fingers. Every failure found in that state is a real defect; every one found sitting still at a desk is a preference.

What is universal, what is learned

Perception is shared. Meaning is not. The system relies on the first and never assumes the second.

UNIVERSAL — SAFE TO BUILD ON

Luminance contrast. Proximity grouping. Motion salience. Closure and continuity. Gravity, mass and inertia. Faces and hands. More-is-up. The 44-pt fingertip. The 8.33 ms frame. Every one of these is a property of the visual system, not of a market.

LEARNED — MUST BE LOCALISED

Colour meaning: red is loss in Western markets and gain in much of East Asia — so Signal hues are always paired with a sign, a word or a position. Reading direction, numeral systems, name order, address shape, date and decimal separators, week start, calendar system, honorifics, and what counts as a polite refusal.

MIRRORING RULES FOR RTL

Mirror layout, navigation direction, progress and chevrons. Do **not** mirror clocks, media transport controls, physical objects, the metal gradient's light source, or any number. The single light source stays at 12° above the viewer in every locale — light does not have a reading direction.

TEXT EXPANSION

German and Finnish run 30–40% longer than English; Japanese and Chinese are shorter but taller. Every capsule is built to grow, never to truncate, and no layout depends on a label's measured width. Combined with AX5 Dynamic Type this is the real stress test of a grid.

This is the practical reason hierarchy is carried by luminance and geometry rather than by colour symbolism.

PROHIBITED PATTERNS

- ✗ **Confirmshaming** — an opt-out worded to embarrass. The refusal is written in the same tone as the acceptance.
- ✗ **Asymmetric weight** — the accepting action wearing metal while the declining one is a grey link. Peer choices are peers.
- ✗ **Manufactured urgency** — countdowns and "3 left" where the constraint is not real.
- ✗ **Roach motel** — signing up in one tap and cancelling in nine. Exit must be as short as entry.
- ✗ **Pre-ticked consent**, hidden cost, infinite feed with no seam, and any streak the user did not ask for.
- ✗ **Fake progress** and artificial delay staged to look like work.

ATTENTION IS THE USER'S, NOT OURS

The system optimises for the user finishing and leaving. Time-in-app is not a success metric here; **task completion, error rate and voluntary return** are. Every technique on sheets 04–12 is powerful enough to be used against the person holding the device, and the only protection is a written rule against it.

THE TEST

Explain the mechanism to the user in one sentence. If they would still choose it, ship it. If the technique requires them not to understand it, it is not a design decision — it is an extraction, and it does not belong in an object built to be inherited.

How every claim is checked

A psychological claim without an instrument is an opinion wearing a citation.

Effect sizes in this field are modest and context-bound. The protocol below exists so the system can be **wrong in public** and corrected.

CLAIM UNDER TEST	INSTRUMENT	PASS THRESHOLD	CADENCE	N
First impression holds	5-second test + 7-point desirability scale	≥ 70% name the primary action unprompted	Per new surface	30
Hierarchy is real	8 px blur ranking + first-click test	Primary action and hero figure ranked 1–2; first click ≥ 80% correct	Per screen	20
Search is pre-attentive	Timed Visual search at 2 densities	Time to find the accent flat as item count doubles	Per release	24
Load fits the ledge	Task success + NASA-TLX + think-aloud	≥ 90% unassisted success; no step over 4 chunks	Per flow	12
Motion feels caused	Frame trace on the slowest supported device + agency probe	P99 ≤ 8.33 ms, zero tasks > 50 ms, feedback < 100 ms	Every build	CI
Legibility in the world	Computed contrast audit + outdoor field read at 400 nits	All text ≥ 4.5:1 against worst-case backdrop; readable in sunlight	Every build + quarterly	CI + 6
Perceived speed	Paired comparison at matched real latency	Skeleton variant preferred ≥ 2 : 1 over spinner	Per loading pattern	40
Accessible under settings	Screen-reader traversal of 12 core flows at AX5, Bold, High Contrast	100% completion, zero truncation, zero unlabelled control	Pre-release gate	manual
Coherence over time	Unbranded recognition test + UMUX-Lite trend	≥ 75% match three screens to one product with labels removed	Twice yearly	60

GUARDRAILS, NOT LIFTS

Experiments here may only be run with completion, error rate and perceived-effort guardrails attached. A variant that raises engagement while raising error rate has failed, whatever the primary metric says.

SMALL N, HONEST CLAIMS

Five participants surface most severe usability defects; they do not measure preference. Qualitative findings are reported as defects, quantitative claims require the sample sizes above, and neither is dressed up as the other.

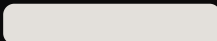
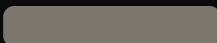
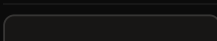
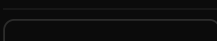
THE SYSTEM IS FALSIFIABLE

Any rule in this document may be overturned by evidence gathered this way — and only by evidence gathered this way: Taste opens the argument; measurement closes it.

Titanium

--ti-00 → --ti-100 · 13 steps · hue 75–85° · chroma ≤ 0.015

The structural ramp. Warm-neutral, brushed rather than bright — a grey with iron in it. Every surface, stroke and label in the system is drawn from these thirteen values and nothing else.

SWATCH	TOKEN	HEX	OKLCH	ON TI-100	ON TI-00	ASSIGNED ROLE
	Titanium White	#F7F6F4	97.3% .003 84.6	18.22	1.00	Light canvas · primary label on dark
	Titanium 05	#EFEDEA	94.7% .005 78.3	16.84	1.08	Grouped-content fill, light
	Titanium 10	#E3E0DB	90.8% .008 80.7	14.95	1.22	Secondary label on dark · light hairline
	Titanium 20	#CFCBC4	84.3% .011 81.8	12.18	1.50	Body copy on dark · brushed-metal mid
	Titanium 30	#B4AFA7	75.6% .013 79.8	9.03	2.02	Tertiary label on dark · inactive glyph
	Titanium 40	#98938A	66.5% .014 82.4	6.44	2.83	Quaternary label on dark · overline caps
	Titanium 50 ♦	#7C776E	57.1% .015 82.4	4.42	4.12	Pivot — the one value legible on both canvases
	Titanium 60	#615D56	48.0% .012 81.8	3.01	6.06	Tertiary label on light · dark-theme metadata
	Titanium 70	#4A4741	39.9% .011 84.6	2.13	8.57	Secondary label on light · dark-theme stroke
	Titanium 80	#35322E	31.9% .008 75.3	1.54	11.80	Dark elevated fill · pressed dark surface
	Titanium 90	#232120	25.0% .004 48.6	1.23	14.84	Dark grouped fill · sheet base
	Titanium 95	#171615	20.1% .003 67.7	1.09	16.73	Dark secondary canvas
	Obsidian 100	#0B0B0B	15.0% .000 89.9	1.00	18.22	Dark canvas · the void. Never pure #000

WHY NOT #000

On OLED, pure black clips the panel to off and the glass edge loses its lensing. Obsidian keeps 5.9% luminance — enough for refraction to survive, invisible as a colour.

CHROMA DISCIPLINE

Chroma never exceeds 0.015. Above that the ramp reads as beige and the metal accent loses its scarcity. The warmth must be felt, never named.

STEP LOGIC

Steps are spaced in perceptual lightness (≈7 L*), not in hex arithmetic. Adjacent steps are always distinguishable and never confusable.

Aurum



--au-05 → --au-90 · 10 steps · hue 67–81° · desert-titanium champagne

The single metal. Used once per screen, on the element of consequence. Its scarcity is the source of its power — a second gold element halves the value of the first.

SWATCH	TOKEN	HEX	OKLCH	ON TI-100	ON TI-00	ASSIGNED ROLE
	Aurum 05 · Vellum	#FBF3E6	96.7% .019 80.1	17.87	1.02	Desert-Gold theme canvas
	Aurum 10 · Bloom	#F6E7CE	93.3% .037 80.6	16.16	1.13	Accent label on dark · specular top of metal
	Aurum 20 · Sand	#EED6AE	88.6% .059 80.4	13.93	1.31	Glass tint on dark · selected-row wash
	Aurum 30 · Champagne ♦	#E3C089	82.5% .082 78.5	11.40	1.60	Dark-theme accent — the signature value
	Aurum 40 · Brass	#D4A96A	76.0% .095 76.1	9.08	2.01	Filled-capsule background on dark
	Aurum 50 · Aurum ♦	#C08F4E	68.4% .102 72.5	6.82	2.67	Brand core — marks, keylines, never body text on light
	Aurum 60 · Bronze	#A5763D	60.2% .094 69.6	4.93	3.70	Large-text accent on light (AA at 24 px+)
	Aurum 70 · Patina ♦	#835C30	50.7% .079 67.7	3.32	5.49	Light-theme accent — AA for body text
	Aurum 80 · Umber	#5E4223	40.3% .060 67.5	2.14	8.53	Pressed accent on light · engraved rule
	Aurum 90 · Ember	#3A2916	29.6% .040 68.3	1.41	12.89	Accent-tinted dark fill · warm glass base

THE ONE-GOLD RULE

One Aurum element per viewport. If a second is needed, the first was not the most important thing on the screen — demote it to Titanium and re-decide.

METAL GRADIENT

```
linear-gradient(168deg,  
#FBF3E6 4%, #E3C089 26%,  
#8E8A82 52%, #C08F4E 78%, #4A4741)
```

FORBIDDEN

No gold on gold. No gold gradient beneath body text. No gold larger than 18% of viewport area. No gold in error, warning or destructive states — ever.

Aurum Chrome

Six stops, one angle, one break. The most-copied and least-understood object in the system.

This is not a gold gradient. It is a **reflection** — a polished surface returning a titanium sky above and a warm ground below, with one specular ridge where the curvature turns.

AURUM

THE SIX STOPS · READ TOP TO BOTTOM ALONG 168°

STOP	POS	HEX	OPTICAL ROLE
Crown	4%	#FFFFFF	Pure specular crest. Not off-white — the one place in the system where #FFF is legal.
Bloom	22%	#F6E7CE	Warm halation around the crest. This single stop is what stops the metal reading as chrome-plated steel.
Body	46%	#8E8A82	Titanium mass. The mid-tone that gives the surface volume; remove it and the gradient becomes a wash.
Break ◆	58%	#F7F6F4	The ridge. A 12-point luminance jump over 12% of the run — the steepest slope, and the whole illusion. It must sit at 54–60%.
Core	76%	#C08F4E	Aurum 50 — the brand value itself, and the only stop drawn from the ramp unmodified.
Root	100%	#4A4741	Shadow terminator. Titanium 70, not brown — the metal is lit by one grey light, so its darkest return is grey.

```
--metal-aurum: linear-gradient(168deg in oklab,
  #FFFFFF 4%, #F6E7CE 22%, #8E8A82 46%,
  #F7F6F4 58%, #C08F4E 76%, #4A4741 100%);
```

WHY 168°

The system has one light source, 12° above the viewer. A vertical gradient is 180°; tipping it back by exactly those 12° puts the crest where that light would actually strike. Every metal surface in the product uses the same angle, so all of them appear lit by the same lamp. **168° is not a taste value — it is the light source written as a number.**

INTERPOLATE IN OKLAB

sRGB interpolation drags the Body→Break transition through a muddy olive. Interpolating in oklab keeps perceptual lightness linear across the ridge, so the highlight reads as light rather than as a colour change. Where the syntax is unsupported, add two intermediate stops at 50% and 54% rather than accepting the sRGB path.

BANDING CONTROL

A six-stop ramp across 900 px on an 8-bit panel bands visibly in the Body region. Two remedies, in order: keep any single run under 40% of the box, and overlay a 0.4-opacity fractal-noise layer at mix-blend-mode: overlay. Never blur the gradient to hide banding — it kills the ridge.

THE GLOW IS SEPARATE

The bloom around the mark is not part of the gradient. It is drop-shadow(0 0 42px rgba(227, 192, 137, 0.28)) — emitted light, zero offset, Aurum 30 at 28%. On light finishes it is removed entirely; warm metal does not glow in daylight.

NEVER Seven or more stops · a rotating angle · animating the angle · two metal objects in one frame · the metal behind text · the metal on a curve or an arc · a coloured drop shadow beneath it.

Five alloys

One per finish. Same angle, same six-stop structure, same break position — only the reflected environment changes.

A metal reflects its surroundings. Change the room and the alloy changes with it — which is why each finish gets exactly one, and no product may mix two.



AURUM

Aurum Chrome *signature*

Black Titanium. The canonical alloy — titanium sky above, warm ground below. Every other alloy is a deviation from this one.

```
#FFFFFF 4%
#F6E7CE 22%
#8E8A82 46%
#F7F6F4 58%
#C08F4E 76%
#4A4741 100%
```

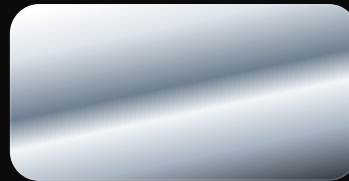


AURUM

Solar

Desert Gold. Gold-dominant, no grey in the body. Ceremonial only — first launch, upgrade, anniversary. Never in a working surface.

```
#FFFD8 4%
#FBE8D2 20%
#D4A96A 44%
#FBE8D2 58%
#A5763D 78%
#3A2916 100%
```

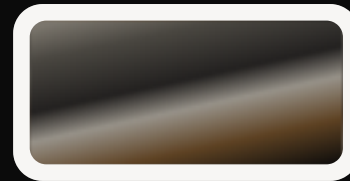


AURUM

Vapor Steel

Vapor Chrome. Gold suspended entirely; the reflected environment is cold sky. For spatial, 3-D and launch experiences.

```
#FFFFFF 4%
#DDE4EA 22%
#6E7F90 46%
#F2F6F9 58%
#A9B6C4 76%
#232B33 100%
```

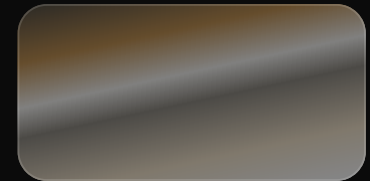


AURUM

Obsidian shown on Ti-00

Natural & White Titanium. The inverse alloy, for metal on a light canvas. Dark returns, one bright ridge — a blackened steel case.

```
#7C776E 4%
#4A4741 20%
#232120 44%
#98938A 58%
#5E4223 78%
#0B0B0B 100%
```



AURUM

Engraved

The signature alloy at 348° — the light source flipped, so the surface reads as cut into the material. Pressed states and etched marks only.

```
348° · same stops
+ inset 0 2 6
  rgba(11,11,11,.5)
no outer glow
never animated
never as fill
```

THE INVARIANTS

Angle 168° (or its 348° inverse). Six stops. Break at 54–60%. Crown within 6% of the start. Root at exactly 100%. Any alloy that violates one of these is not part of the family.

HOW A NEW ALLOY IS DERIVED

Take the reflected environment of the finish — its canvas and its accent — and substitute at Bloom, Body, Core and Root. Crown and Break stay near-white in every alloy, because a specular highlight is the colour of the light, not of the metal.

FOIL EQUIVALENTS

In print the gradient is not reproduced — it is replaced. Aurum Chrome → cold-foil silver with a warm-gold overprint at 22%; Solar → gold foil, matte; Obsidian → black foil on uncoated stock. Never a CMYK simulation of a metal.

The wordmark

Set, not drawn. The mark is SF Pro Display Bold at -0.055 em with the signature alloy — nothing is redrawn by hand except the kern pairs.

A mark that is set from the system's own typeface cannot drift away from the system. That is the entire reason it is not a logo.



CLEAR SPACE

Half the cap height on every side, measured from the glyph extents — not from a bounding box. Nothing enters it, including a page edge.

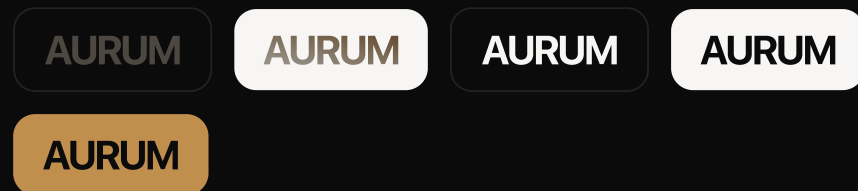
MINIMUM SIZE

Metal alloy: **28 px** cap height. Below that the ridge falls inside one pixel row and the mark reads as dirty. Solid variants: 14 px.

KERNING

Above 48 px, AU -8 , RU -4 , UM -6 units. Metric kerning is engineered for 17 px and always reads loose on a wall.

PERMITTED VARIANTS, IN ORDER OF PREFERENCE



Metal on obsidian is the primary. Obsidian metal on paper-white is the light equivalent. The two solid variants exist for one-colour reproduction, favicons, embroidery and any surface under 28 px. Knockout on Aurum 50 is for stamps and stickers only.

MISUSE

- ✗ Rotating or arcing the mark. The alloy is lit by a fixed source; rotate the mark and the light rotates with it, which no real metal does.
- ✗ Outlining or stroking. A stroke flattens the ridge into a shape and the surface becomes a sticker.
- ✗ Placing it on imagery or on glass. Metal needs a plain dark or plain light ground; a busy backdrop destroys the read of the highlight.
- ✗ Stretching, condensing or re-weighting. The mark has one weight and one width. If it does not fit, the layout is wrong.
- ✗ Adding a bevel, emboss, inner shadow or second highlight. The gradient already contains the entire lighting model.
- ✗ Two marks in one view, or the mark beside another metal object. Scarcity is the value.
- ✗ Recolouring the alloy to match a campaign, a season or a partner brand.

LOCKUP WITH THE DESCRIPTOR



Descriptor is Overline — 10/12, $+0.20$ em, Titanium 40, centred under a hairline whose width is 40% of the mark. Gap above the rule is $\frac{1}{4}$ cap height, below it $\frac{1}{4}$ cap height. The descriptor never appears without the mark, and never in metal.

Applying the metal

Four legal surfaces, one animation, one hard accessibility limit.

Total metal area across a whole product should be under **1%** of pixels shipped. It is a punctuation mark, not a finish.

27.0

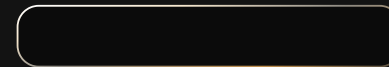
Type fill

Display steps only — 34 px and above, weight 700. One string per screen. Never a sentence, never a label, never a number the user must read to act.

Launch

Surface fill

The three-stop working reduction — Bloom, Core, Root — for capsules. The full six-stop alloy is never used on a control; a button is a surface, not a jewel.



Hairline stroke

1 pt gradient border via a padded parent. For the single most important card on a screen. Radius follows the concentric rule: inner = outer – 1.



The object

App mark, medal, seal. Squircle at 158° — a physical object is a slightly different plane from flat type. Carries the e5 glow. Once per product.

THE SWEEP — THE ONLY PERMITTED ANIMATION

```
@keyframes aurum-sweep {
  from { background-position: 0% 0%; }
  to   { background-position: 0% 200%; }
}

.mark {
  background-size: 100% 200%;
  animation: aurum-sweep 2400ms
    cubic-bezier(.16,1,.3,1) 1 both;
}
```

The gradient is twice the box height and travels once, on the Enter curve, over 2400 ms — the ridge crossing the glyphs exactly once as if a light passed over the object. It runs on **first paint only**: never on hover, never on scroll, never on a loop. A metal that shimmers continuously is a hologram, and a hologram is the opposite of provenance. Under `prefers-reduced-motion` the mark simply appears at its rest position.

PERFORMANCE

Animating `background-position` on a text-clipped element repaints every frame. The mark is therefore promoted for the duration of the sweep and demoted on completion, and it is the only element on screen animating during first paint — the 8.33 ms budget is spent entirely on it.

THE ACCESSIBILITY LIMIT

The Root stop is Titanium 70 — 2.13:1 on obsidian. Gradient type therefore **cannot meet the text floor** and is governed by three absolute rules: it appears only at Display sizes where the large-text 3:1 threshold applies; the Root stop terminates below the baseline so no letterform interior sits inside it; and the string is always duplicated as accessible text elsewhere on the surface, or is a proper noun carrying no information. Under Increase Contrast the alloy is replaced by `--content-primary`, flat.

ON GLASS

Metal never sits on glass and glass never sits on metal. Both are specular materials with their own lighting model; stacked, they contradict each other and the whole illusion of a single light source collapses.

SCREENSHOTS & EXPORT

The alloy must be rasterised at 2× or above and exported as PNG. JPEG artefacts collect along the ridge, and any conversion that quantises to 8-bit per channel needs the noise overlay reapplied afterwards.

AUDIT Count metal instances per build. One per screen, one glowing object per product, under 1% of shipped pixels. If the count rises, the metal is no longer a signal — it is a texture, and the value is already gone.

Vapor & Signal

VAPOR — THE COLD AXIS

Two restricted sets. Vapor cools glass and distance; Signal names state. Neither is ever used for brand expression or decoration.

hue 244–250° · chroma ≤ 0.033 · for glass tint, depth & atmosphere only

Vapor 10

#DDE4EA · 15.33
Specular rim on cold glass

Vapor 30

#A9B6C4 · 9.54
Distant-object label

Vapor 50

#6E7F90 · 4.78
Atmospheric mid-depth

Vapor 70

#44505C · 2.39
Cool elevated fill

Vapor 90

#232B33 · 1.37
Chrome-theme canvas

SIGNAL — RESERVED FOR STATE

four hues · one meaning each · never brand, never decoration

Nominal

#30D158 dark · #248A3D light
Success, completion, system-within-tolerance. 9.74:1 on obsidian.

Caution

#FFD60A dark · #9A6C00 light
Degraded, approaching limit. Never paired with gold. 13.94:1.

Abort

#FF453A dark · #D70015 light
Failure and destruction. Always accompanied by an action. 5.78:1.

Link

#0A84FF dark · #0064D2 light
Navigation away from the current context only. 5.40:1.

NEVER COLOUR ALONE

Every Signal use carries a redundant channel — an SF Symbol, a position or a word. Eight per cent of men cannot resolve Nominal from Abort by hue. A system that fails them fails the law.

TOTAL COLOUR BUDGET

Thirteen Titanium + ten Aurum + five Vapor + four Signal = 32 values. A screen may use at most nine of them. No value outside this set may enter the codebase — including imported illustration and third-party embeds.

Five finishes

The same 32 values, five assignments. A product picks one and never mixes. Each finish is a complete light-or-dark pair; the two Titanium finishes are the defaults.

Launch

Later

Black Titanium
DEFAULT · DARK

Cinema, media, telemetry, night. The canonical AURUM expression — the void with one warm light in it.

Launch

Later

Natural Titanium
DEFAULT · LIGHT

Documents, settings, commerce, daylight. Brushed paper-white with an engraved dark label.

Launch

Later

Desert Gold
CEREMONIAL · LIGHT

Onboarding, upgrade, receipts, anniversaries. The one finish where gold may be the ground.

Launch

Later

White Titanium
HIGH-KEY · LIGHT

Editorial, gallery, long-form reading. No accent at all — hierarchy carried purely by type weight.

Launch

Later

Vapor Chrome
IMMERSIVE · DARK

Spatial, 3-D, launch experiences. Gold is suspended; cold light carries the whole hierarchy.

SWITCHING

A theme change is a 320 ms cross-fade of the token layer only — geometry, type and motion are identical across all five, so no layout recalculates and nothing shifts by a single pixel. Finish is a costume; the skeleton never changes.

The token layer

Product code never references a ramp value. It references a role. This is the only table an engineer needs.

42 roles

full parity

TOKEN	BLACK TITANIUM	NATURAL TITANIUM	RULE OF USE
SURFACE			
--surface-canvas	ti-100	ti-00	The floor. Never elevated, never tinted, never glass.
--surface-grouped	ti-95	ti-05	Behind lists and forms; inverts the canvas relationship.
--surface-raised	ti-90	ti-00	Cards. One level only — no card inside a card.
--surface-glass	ti-00 @ 9%	ti-00 @ 62%	Floating chrome only; bars, popovers, sheets. Sheet 25.
--surface-scrim	ti-100 @ 56%	ti-100 @ 34%	Behind modals; always animates with the sheet, never alone.
CONTENT			
--content-primary	ti-00	ti-100	Titles and the sentence that matters. 18.22:1 both themes.
--content-secondary	ti-20	ti-70	Body. 12.18 / 8.57. The workhorse of every screen.
--content-tertiary	ti-40	ti-60	Metadata, timestamps, captions. 6.44 / 6.06 — still AA.
--content-quaternary	ti-60	ti-40	Placeholder and disabled only. Never a real word.
--content-accent	au-30	au-70	One instance per viewport. 11.40 / 5.49.
--content-on-accent	ti-100	ti-00	Label on a filled metal capsule. Obsidian on gold, always.
STROKE & FILL			
--stroke-hairline	ti-00 @ 9%	ti-100 @ 9%	0.5 pt separators. Prefer a gap; use a rule only when space fails.
--stroke-rim	ti-00 @ 18%	ti-100 @ 12%	The 1 pt edge of every glass object. Non-optional.
--stroke-focus	au-30, 3 pt	au-70, 3 pt	2 pt offset outside the shape. Never removed, never inset.
--fill-quiet	ti-00 @ 7%	ti-100 @ 5%	Secondary capsule ground, segmented track, chip.
--fill-pressed	ti-00 @ 14%	ti-100 @ 11%	Applied with scale 0.96 — colour alone never reads as pressed.
--fill-accent	au-30 → au-40	au-50 → au-70	168° two-stop gradient, 6% delta. Flat metal looks printed.
LIGHT			
--specular-top	#FFF @ 45%	#FFF @ 80%	1 pt inset highlight, top edge only. The tell of real glass.
--specular-bottom	#FFF @ 12%	ti-100 @ 6%	Bounce light. Always weaker than the top by ≥3×.
--shadow-key	ti-100 @ 60%	ti-100 @ 14%	Single source, 12° above. Sheet 32 defines the ladder.
--glow-accent	au-30 @ 28%	none	Dark themes only, 42 px blur, zero offset. Never on text.

ONE DIRECTION ONLY	OPACITY OVER NEW HEX	NAMING	GOVERNANCE
Roles reference ramp values. Ramp values reference nothing. A role that references another role is an alias, and aliases are audited out at every release.	Strokes and fills are stated as a ramp value at a percentage, never as a new opaque colour — so they stay correct over glass, over media and in both themes.	category--role--state. No product prefixes, no screen names, no --new, no --v2. A token that needs a version is a token that should have been retired.	Additive only. A token's value is never edited in place — it is deprecated, replaced and removed one release later, so no product ever changes appearance without a commit.

Legibility matrix

Measured WCAG 2.2 ratios. Every pairing shipped in AURUM appears here; anything absent is forbidden.

AAA ≥ 7.0 AA ≥ 4.5 Large-only ≥ 3.0 Non-text only

ON OBSIDIAN · #0B0B0B

Titanium White — primary label	18.22 AAA
Titanium 20 — body copy	12.18 AAA
Aurum 30 — accent label	11.40 AAA
Titanium 30 — tertiary	9.03 AAA
Aurum 50 — mark & keyline	6.82 AA
Titanium 40 — caption	6.44 AA
Titanium 60 — disabled	3.01 non-text

Titanium 60 and below are structural on dark: strokes, tracks, dividers and disabled glyphs. They may never carry a word a user must read.

ON TITANIUM WHITE · #F7F6F4

Obsidian — primary label	18.22 AAA
Titanium 90 — display	14.84 AAA
Titanium 70 — body copy	8.57 AAA
Titanium 60 — tertiary	6.06 AA
Aurum 70 — accent label	5.49 AA
Titanium 50 — caption	4.12 AA·large
Aurum 50 — never as text	2.67 forbidden

The gold that carries the brand on dark cannot carry a word on light. Light themes step two rungs down the ramp to Aurum 70 — the same metal, a legible value.

FLOOR

4.5:1 for every readable string, 3:1 for icons and control edges, no exceptions granted for aesthetic reasons.

GLASS CAVEAT

Text on glass is measured against the **worst-case** backdrop, not the average. If it fails, the glass gains an adaptive tint — the text never darkens.

HIGH CONTRAST

Increase Contrast promotes every label one rung, doubles rim opacity, and replaces glass with an opaque fill.

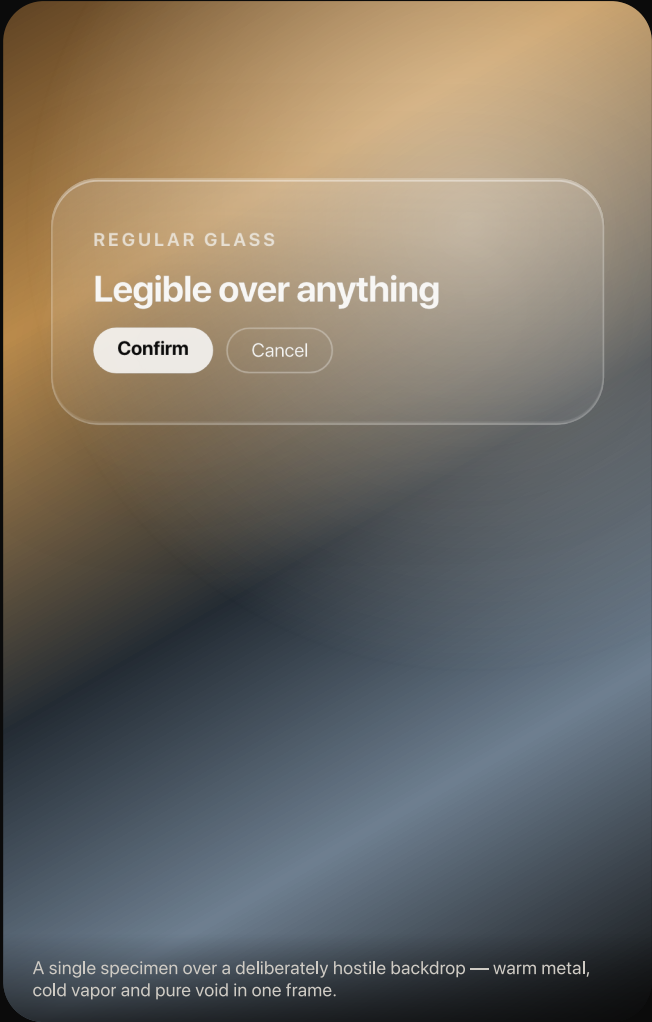
AUDIT

Contrast is asserted in CI against the token table. A build that introduces an unlisted pairing fails.

Liquid Glass

Eight layers, composited in order. Omit any one and the material reads as a translucent rectangle.

Glass refracts what is beneath it, reflects what is around it, and lenses along its edge. It is a lens — not a filter.

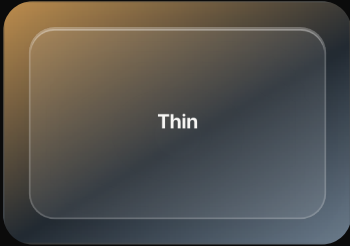
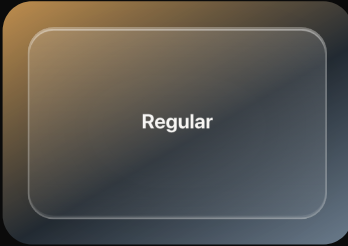
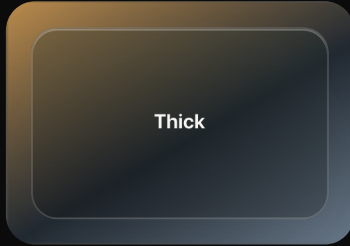
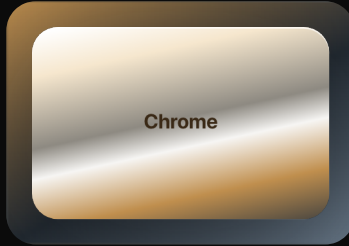
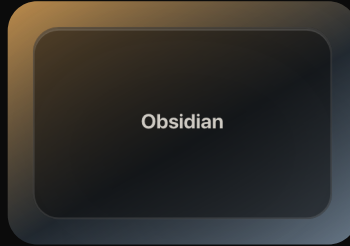


01	Backdrop capture <code>isolation: isolate</code>	The content below is sampled live at device resolution. Never a static blur, never a screenshot — glass that does not update on scroll is dead glass.
02	Edge lensing <code>displace 12 px · n ≈ 1.45</code>	The outer 12 px displaces the backdrop inward and magnifies it ~4%. This single detail is what separates glass from frosted plastic.
03	Blur <code>blur(20 / 28 / 40 px)</code>	Thin, regular and thick. Blur radius scales with the object's implied thickness, never with its size.
04	Saturation lift <code>saturate(180%)</code>	Blur desaturates. Restoring chroma keeps the colour of the world visible through the object and stops glass turning grey.
05	Luminance adaptation <code>8–14% auto tint</code>	The object measures the backdrop's mean luminance each frame and adds a light or dark wash to hold its label above 4.5:1. Text never changes colour; the glass does.
06	Tint <code>ti / au / az @ 4–10%</code>	Optional identity wash. Above 10% the material stops being glass and becomes a coloured panel.
07	Specular rim <code>inset 0 1px #FFF/45%</code>	One bright pixel on the top edge, one dim pixel on the bottom, 1 pt rim all round. The highlight follows the single light source and never rotates.
08	Contact shadow <code>0 18px 40px / 45%</code>	Proves the object is above the content rather than cut into it. Scales with elevation; see sheet 32.

NEVER Glass on glass · glass over a plain flat fill (there is nothing to refract) · glass behind long-form body text · more than three glass planes in one frame · glass with a coloured drop shadow.

Five glasses

Copy-exact values. A product may add none. Each specimen below sits on the same backdrop so the difference is the material, not the scene.

				
Thin	Regular default	Thick	Chrome	Obsidian
Tab bars, toolbars, always-present chrome. Reads as a veil, not an object.	Sheets, popovers, alerts, cards that float. The workhorse — 80% of all glass.	Full-height sheets and modals that must fully suppress the scene behind them.	Marks, medals, the single hero object. Opaque, not translucent — polished metal.	Video and camera overlays, immersive HUDs. Holds white text over any imagery.
<pre>blur(20px) saturate(170%) bg #F7F6F4/10% rim /20% · 1pt spec /40% shadow 0 8 20 /35%</pre>	<pre>blur(28px) saturate(180%) bg /16% → /6% rim /26% · 1pt spec /45% + /10% shadow 0 14 32 /45%</pre>	<pre>blur(40px) saturate(150%) brightness(0.86) rim /18% · 1pt spec /28% shadow 0 20 44 /55%</pre>	<pre>no backdrop-filter 168° 6-stop metal spec /70% shadow 0 14 34 /50% max 1 per screen never behind text</pre>	<pre>blur(32px) brightness(0.34) bg #0B0B0B/52% rim /12% · 1pt spec /16% shadow 0 16 38 /60%</pre>

PERFORMANCE

Each glass plane costs one full-screen GPU pass. Three planes is the hard ceiling on mobile; above that the 8.33 ms budget breaks and the material's own promise of smoothness is broken with it.

DEGRADATION

Where `backdrop-filter` is unavailable or Reduce Transparency is on, each recipe collapses to a flat token fill at the same luminance — surface-raised for Thin and Regular, ti-90 for Thick, ti-100/92% for Obsidian.

One voice, three cuts

SF Pro is the only typeface in the system. No display face, no serif, no third-party licence, no exceptions.

A neutral grotesque with humanist apertures — legible at 10 pt on a watch and monumental at 96 pt on a wall.

Aa

SF Pro Display

≥ 20 pt · weights 400 600 700

Tighter apertures, smaller x-height ratio, negative tracking. Every title, every number that matters, every word set larger than the body.

Aa

SF Pro Text

< 20 pt · weights 400 500 600

Wider spacing, larger apertures, taller x-height. Body copy, labels, controls — everything a user reads rather than notices.

A0

SF Mono

any size · weights 400 500 600

Telemetry, identifiers, timestamps, code, token names. The Frontier axis made audible. Never used for prose.

OPTICAL SIZING IS MANDATORY

The cut switches at **20 pt**, not by eye. Display below 20 pt collapses its counters; Text above 20 pt looks slack and amateur. On the web this is `font-optical-sizing: auto` plus an explicit family switch at the breakpoint.

THREE WEIGHTS, ONE JOB EACH

Regular reads. **Semibold** labels and emphasises. **Bold** is reserved for Display steps only. Light, Thin, Ultralight and Black are not licensed for product use — thin type on glass is illegible and thin type at scale is a trend.

MEASURE

45–75 characters. Below 45 the eye ricochets; above 75 it loses the return. Long-form uses `text-wrap: pretty`; titles use `text-wrap: balance`. Never justify, never hyphenate outside CSS columns.

TRACKING FOLLOWS SIZE

Letter-spacing is a function of size, never a taste decision: -0.045 em at 96 pt easing to 0 at 16 pt and reopening to $+0.16\text{ em}$ for 10 pt overlines. Sheet 28 fixes every value.

NUMERALS

Proportional in prose. **Tabular** — `font-variant-numeric: tabular-nums` — in every table, timer, counter and price, so digits never jitter as they change. A live number that reflows its neighbours destroys the illusion of an instrument.

FALLBACK STACK

```
font-family: "SF Pro Display",
  -apple-system, BlinkMacSystemFont,
  "Google Sans", "Helvetica Neue",
  system-ui, sans-serif;
```

DYNAMIC TYPE

Every step scales with the platform text-size setting from xSmall to AX5 — a 310% range. Layouts are built so that at AX5 rows stack vertically rather than truncate; no label is ever clipped and no capsule ever loses its 44-pt target. Type that cannot grow is not accessible, and a system that is not accessible is not premium.

Fourteen steps

A closed set. If a design needs a fifteenth size, the design is wrong.

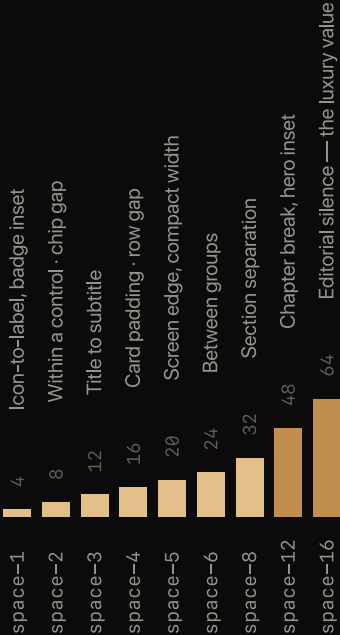
Leading is expressed in points, not multiples, so every step lands on the 4-pt grid.

STEP	SIZE	LEADING	TRACKING	WEIGHT	CUT	WHERE IT IS ALLOWED
Colossus	96	88	−0.045	700	Display	One per product. Launch screen, hero metric, wordmark.
Display 1	64	64	−0.038	700	Display	Section openers, marketing headline, empty-state title.
Display 2	48	52	−0.032	700	Display	Large-title navigation bar, onboarding step title.
Title 1	34	40	−0.026	700	Display	Screen title, sheet title, card headline.
Title 2	28	34	−0.021	600	Display	Group heading, modal title, statistic label.
Title 3	22	28	−0.016	600	Display	Subsection heading, list-section header.
Headline	17	22	−0.010	600	Text	Row title, button label, active tab, emphasis in body.
Body	17	25	−0.004	400	Text	The default. Every sentence a user actually reads.
Callout	16	21	0	400	Text	Secondary paragraph, inline notice, quoted line.
Subhead	15	20	0	400	Text	Row subtitle, supporting line under a title.
Footnote	13	18	+0.005	400	Text	Legal, disclaimer, secondary metadata.
Caption	12	16	+0.010	400	Text	Image caption, timestamp, tab-bar label.
Micro	11	13	+0.014	500	Text	Badge count, chart axis, dense telemetry.
Overline	10	12	+0.160	600	Text · caps	Eyebrow above a title. Never more than four words.

Four, and multiples of four

One base unit for the whole system. Every dimension in every product is 4 × n.

THE SPACING LADDER



Six, ten, eighteen and twenty-two do not exist. A designer who needs 18 px is compensating for a mistake two levels up the tree.

PROXIMITY BEATS DIVISION

Related items sit at **space-2**, unrelated items at **space-6** or more. That ratio of 1 : 3 is enough for the eye to group without a single line being drawn. A divider is admitted only when two groups must touch.

Rhythm is the cheapest form of fluency. A grid the user never sees is doing the most work on the page.

LAYOUT GRIDS

CLASS	WIDTH	COLS	GUTTER	MARGIN
Compact	≤ 430	4	16	20
Medium	431–833	8	20	32
Regular	834–1279	12	24	48
Expanded	1280–1919	12	24	64
Cinema	≥ 1920	12	32	auto

Content is capped at **1440 px** and centred beyond Cinema. A line of text that spans a 34-inch display is a failure of nerve, not a feature of the display.

SAFE AREA & REACH

TOP	Never place interactive content within 8 pt of the sensor housing; the status bar is glass, not padding.
BOTTOM	34 pt home indicator + 8 pt clearance. Primary actions live in the bottom third — the thumb arc, not the visual centre.
TARGET	44 × 44 pt absolute minimum, 48 preferred, 8 pt minimum separation between adjacent targets.
DESTRUCT	Destructive actions sit ≥ 24 pt from any confirming action and never in the thumb arc.

The Capsule Law

Anything a finger touches is a capsule. Anything that holds content is a squircle. Nothing is a rectangle.

A capsule has no corners to resolve, so it reads as a single object in one saccade — the fastest shape the eye can parse.

Capsule

Squircle 20

Squircle 28

Squircle 12

Rectangle

$99\text{px} \cdot h \leq 56$

radius-lg

Buttons, chips, toggles, tabs, search, badges, avatars.

radius-xl

Sheets, modals, full-width surfaces.

radius-sm

Inline chips, code blocks, swatches, thumbnails.

forbidden

Only full-bleed edges that reach the screen boundary may be square.

CONTINUOUS CURVATURE

A circular corner joins its straight edge with a curvature discontinuity the eye registers as a faint crease. A **squircle** — superellipse, exponent ≈ 5 — eases curvature to zero and the join disappears. Every radius in AURUM is continuous.

```
/* native */
corner-shape: squircle;
border-radius: 20px;

/* fallback */
border-radius: 20px; /* +8% */
```

Where the superellipse is unavailable, a circular radius **8% larger** matches the perceived roundness.

CAPSULE SIZING

SIZE	H	PAD-X	LABEL
XS	28	12	12
S	36	16	15
M default	44	20	17
L	52	24	17
XL	64	32	22

Above 56 pt height the pill silhouette becomes a lozenge and loses its authority — switch to **radius-xl**. Icon-only capsules are perfect circles at the same heights.

CONCENTRIC RADII

A child inside a rounded parent takes the parent's radius **minus the gap** between them. Never the same radius, never an arbitrary one — mismatched concentricity is the single most common tell of amateur work.



$r\text{-inner} = r\text{-outer} - \text{gap}$
clamp at radius-sm (12) minimum
0 only where the child bleeds

One light, six heights

A single source, 12° above the viewer, infinitely far away. It never moves, never colours, never multiplies.

Two light sources in one interface is the visual equivalent of two vanishing points — the brain notices immediately and trusts nothing after.

e0	e1	e2	e3	e4	e5
Flush no shadow 1pt rim only List rows, grouped fills, inline content.	Resting 0 1 2 / 40% 0 0 1 / 24% Cards, tiles, quiet capsules.	Lifted 0 4 10 / 46% 0 1 2 / 28% Hover, dragged item, filled capsule.	Floating 0 10 24 / 52% 0 2 5 / 30% Toolbars, tab bars, popovers.	Sheet 0 13 44 / 60% 0 4 10 / 34% Modal sheets, side panels, menus.	Singular 0 28 70 / 70% + glow 42 au / 20% The one hero object. Once per product.

TWO SHADOWS, ALWAYS

Every elevation is an **ambient** shadow — wide, soft, offset by the elevation — plus a **contact** shadow: tight, dark, offset by a quarter. A single shadow reads as a sticker; two read as an object resting in a room.

BLUR IS 2.4 × OFFSET

The ratio is fixed across the ladder, so an object at any height belongs to the same physical scene. Spread stays at zero — a spread value is a designer trying to fake a second light.

LIGHT THEMES HALVE EVERYTHING

On Titanium White, opacity drops to roughly a quarter of the dark values (e3 becomes 0 10 24 rgba(11,11,11,0.13)). A dark-theme shadow on a light canvas looks like dirt.

ELEVATION IS ANIMATED BY OPACITY

Never animate box-shadow — it repaints every frame. Render both shadow states as stacked pseudo-elements and cross-fade their opacity on the compositor. See sheet 35.

THE GLOW IS NOT A SHADOW

e5's Aurum bloom is emitted light, so it has zero offset and sits **behind** the object with no contact pair. It appears on exactly one element per product and never on text, never on a scrollable item, never on more than one at a time.

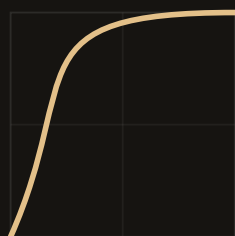
DEPTH ORDER

canvas → grouped → raised → glass chrome → sheet → alert → singular. An object may never sit between two levels, and no product may add a level.

Six curves, six springs

The entire motion vocabulary of the system. Nothing eases linearly; nothing eases arbitrarily.

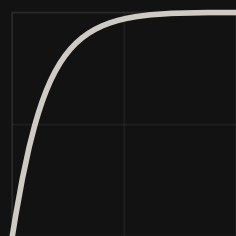
Physical objects begin slowly, accelerate under force and decelerate against resistance. An interface that ignores this reads as a slideshow, however fast it runs.



Standard

```
cubic-bezier(
  0.32, 0.72, 0, 1)
400 ms
```

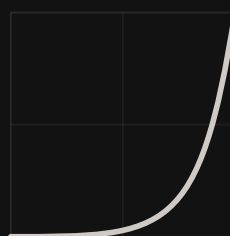
The default for everything that changes position or size. Fast commitment, long graceful settle.



Enter

```
cubic-bezier(
  0.16, 1, 0.3, 1)
520 ms
```

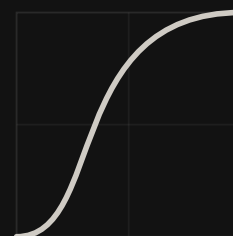
Objects arriving. Near-instant departure from rest, then a very long tail — the shape of something heavy being placed.



Exit

```
cubic-bezier(
  0.7, 0, 0.84, 0)
240 ms
```

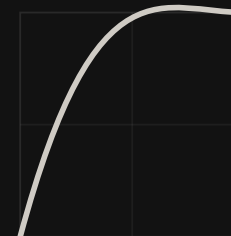
Objects leaving. Departures are always faster than arrivals — the user has already decided.



Glide

```
cubic-bezier(
  0.4, 0, 0.2, 1)
300 ms
```

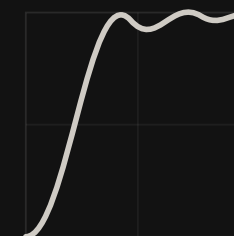
Symmetric in-out for things that travel a known distance: carousels, page indicators, sliding tabs.



Snap

```
cubic-bezier(
  0.34, 1.26, 0.64, 1)
280 ms
```

A 4% overshoot for confirmations and toggles. Used sparingly — overshoot everywhere reads as a toy.



Inertia

```
spring(1, 220, 26)
≈ 380 ms settle
```

Not a duration — a simulation. Any gesture-driven object hands its velocity to this spring on release.

SPRING DEFINITIONS

SPRING	MASS	STIFF	DAMP	SETTLE	APPLIED TO
spring-tap	1	500	32	240 ms	Press & release of any capsule
spring-snap	1	420	26	320 ms	Switch thumb, checkmark, segmented pill
spring-sheet	1	240	28	480 ms	Sheets, drawers, sidebars
spring-morph	1	180	22	620 ms	Shared-element hero transitions
spring-scrub	1	300	34.6	400 ms	Critically damped — sliders, scrubbers, no overshoot
spring-inertia	1	220	26	380 ms	Gesture release; initial velocity from the finger

Damping ratio $\zeta = \text{damping} \div 2\sqrt{(\text{stiffness} \times \text{mass})}$. Below 0.7 the object wobbles and reads as plastic; at 1.0 it is dead and reads as a mechanism. AURUM lives at **0.72–0.85** — the range where an object feels machined from metal.

ANTICIPATION

Before any transition over 400 ms, the origin object compresses by 2% for 120 ms. The eye reads intent before travel begins — the same trick a diver uses on a board, and the reason a 620 ms morph never feels slow.

FOLLOW-THROUGH

Children lag their parent by 24 ms per depth level, capped at three levels. A card and its label arriving on the same frame look printed; 24 ms apart they look attached.

ASYMMETRY IS THE RULE

Enter : Exit is always about 2 : 1. Reversing a transition at its own speed feels like an undo; reversing it faster feels like an escape — and users always want the escape.

INTERRUPTIBILITY

Every animation can be reversed mid-flight from its current position and velocity. An animation that must finish before the next input is accepted is a bug, not a flourish.

Every transition, specified

Twenty-two interactions. If a behaviour is not on this sheet, it does not animate.

Properties listed are the only ones that may change.
Anything else is a performance defect.

INTERACTION	ANIMATED PROPERTY	DURATION	CURVE	STAGGER	HAPTIC
TOUCH					
Press down	scale 0.96 · fill-pressed	90 ms	Exit	—	light · on contact
Release	scale 1	240 ms	spring-tap	—	—
Hover in	opacity of e2 layer	160 ms	Glide	—	—
Toggle	translateX · track fill	320 ms	spring-snap	—	rigid · at 60%
Drag	translate 1:1 · scale 1.03	finger	none	—	light · on lift
Drag release	translate · scale 1	380 ms	spring-inertia	—	medium · on land
NAVIGATION					
Push forward	translateX 100→0 · parent -30%	400 ms	Standard	—	—
Pop back	translateX 0→100	280 ms	Exit	—	—
Sheet present	translateY · scrim opacity · parent 0.94	480 ms	spring-sheet	—	soft · at 40%
Sheet dismiss	translateY · scrim opacity	280 ms	Exit	—	—
Hero morph	transform · border-radius · clip	620 ms	spring-morph	—	—
Tab switch	pill translateX · glyph opacity	300 ms	Glide	—	light
CONTENT					
List appear	opacity 0→1 · translateY 12→0	520 ms	Enter	24 ms ×8	—
Row insert	height · opacity · neighbours translateY	400 ms	Standard	—	light
Row delete	translateX -100% · height 0	240 ms	Exit	—	rigid
Value change	digit roll translateY	280 ms	Snap	16 ms	—
Skeleton → content	opacity cross-fade	240 ms	Glide	24 ms	—
Pull to refresh	translateY rubber-band · arc	finger	spring-inertia	—	medium · at trigger
FEEDBACK					
Success	check stroke-dashoffset · scale	520 ms	Snap	—	success · at 70%
Error	translateX #6 ×2	320 ms	Glide	—	error · on first pass
Toaster enter	translateY · scale 0.94→1	520 ms	Enter	—	soft
Focusing	outline opacity · scale 1.04→1	160 ms	Glide	—	—
Ambient idle	glow opacity 0.18→0.28	4 000 ms	Glide, alternate	—	—

The 8.33 ms contract

120 Hz is not a target. It is the unit of measurement for whether a feature is finished.

FRAME BUDGET



Style 1.2 Layout 1.5 Paint 1.8 Composite 1.0 Reserve 2.8

The reserve is not spare capacity — it absorbs thermal throttling, a background fetch and a garbage-collection pause. A feature that consumes it on a bench will drop frames on a warm device in a user's hand.

COMPOSITOR-ONLY PROPERTIES

ANIMATE

transform
opacity
filter*
clip-path*
backdrop-filter*

* GPU-bound; budget one per frame

NEVER ANIMATE

width / height
top / left / margin
box-shadow
border-radius†
background-position

† except in a hero morph

BATCH

All geometry reads happen before any style write. One interleaved read forces synchronous layout and costs 4–11 ms — more than the whole frame.

GLASS ≤ 3

Each backdrop filter is a full-screen sample-blur-composite pass. Three fits inside 1.0 ms of compositing; four does not, on any device, ever.

A single dropped frame in a 400 ms transition is a 2.4% error — and the human visual system detects it every time.

PROMOTE DELIBERATELY

will-change is applied on gesture start and removed on settle — never left in a stylesheet. A permanently promoted layer consumes GPU memory forever and pushes the very frames it was meant to save.

GESTURE ON THE MAIN THREAD IS FORBIDDEN

Scroll, drag and scrub are driven by the compositor or a worklet. If a gesture's smoothness depends on JavaScript, it will stutter the first time a network response lands mid-swipe.

MEASURED, NOT ASSERTED

Every release ships a frame-timing trace of the twelve core flows on the slowest supported device.

P99 frame time ≤ 8.33 ms, zero long tasks over 50 ms. A regression blocks the release. Smoothness is the only feature that cannot be added later.

REDUCE MOTION

Under prefers-reduced-motion, travel and scale are replaced by a 160 ms opacity cross-fade at the destination. Nothing is disabled and no state is lost — the choreography simply stops moving. Parallax, auto-play and the ambient idle loop are switched off entirely.

The capsule family

Five variants, five states, five sizes. Exactly one Metal capsule may exist on a screen.

h 44 · pad-x 24 · r 999
label 17/600 · ti-100
fill au-30>au-50 168°
spec /55% · e2 · glow 32

	REST	HOVER	PRESSED	FOCUS	DISABLED
Metal	Launch	Launch	Launch	Launch	Launch
Solid	Confirm	Confirm	Confirm	Confirm	Confirm
Glass	Details	Details	Details	Details	Details
Tinted	Add	Add	Add	Add	Add
Plain	Skip	Skip	Skip	Skip	Skip
Destruct	Delete	Delete	Delete	Delete	Delete

COMPOSITION

icon-only capsules are perfect circles at the same five heights. A split capsule's divider is 1 pt at rim opacity. A loading capsule keeps its exact rest width.

XS

S

Medium

Large

XL

☐ Leading icon

Export

Members 12

Hierarchy. One Metal, any number of Glass, one Destruct. Two primaries on one screen means the decision was not made.

Loading. The label is replaced in place by a 16 pt spinner; the capsule never changes width. A resizing button relocates the user's finger.

Labels. A verb the user would say aloud. Never "OK", "Submit", or "Yes" — always "Launch", "Delete draft", "Send \$240".

Icons. Leading icon at 0.82 × label size, optically centred, 6 pt gap. Trailing icons are for disclosure only.

Nine controls

SEGMENTED

Day

Week

Year

Two to four options, all visible, all short. The selected pill translates on the Glide curve — it never fades. Above four options, use a menu.

SWITCH

Thumb travels on spring-snap; the track fill cross-fades over 200 ms so colour arrives **after** the thumb commits. Haptic fires at 60% of travel, not on touch.

SLIDER

062100

Track 6 pt, thumb 24 pt, target 44 pt. Value follows the finger 1 : 1 with no easing; on release it settles on spring-scrub — critically damped, because a scrubber that overshoots is a scrubber that lies.

Every control is a capsule or contained by one. Every control has a 44-pt target regardless of its drawn height.

TEXT FIELD

Callsign

Falcon

Fal · 3 characters minimum

Callsign

Falcon

Fal · 3 characters minimum

Label above, never inside — a placeholder that vanishes destroys the user's only reference. Validation on blur, never per keystroke.

SEARCH

Search 412 tokens

#K

Always Thin glass — search floats above content by definition. Results appear beneath with a 24 ms stagger, capped at eight rows before batching.

SELECTION

☒ On

☐ Off

☐ Radio

Squirrel for multi-select, circle for exclusive — the shape carries the semantics before the label is read. The checkmark draws its stroke over 240 ms; it never pops in.

PROGRESS

74.2% · 00:41 remaining

Determinate whenever a number exists — an indeterminate spinner is a confession. Below 1 s, show nothing at all; a flash of a spinner reads as a fault.

LIST ROW

Natural Titanium

Default · light

Vapor Chrome

Immersive · dark

Rows are 44 pt minimum with a 14-pt inset; the headline starts after the leading element, never at the edge. Chevrons only where a push follows.

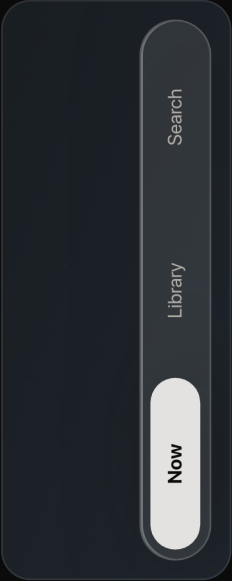
Surfaces & navigation

NAVIGATION BAR



The large title collapses to the compact 17/600 title over the first 44 pt of scroll, driven by scroll position rather than a threshold — so the transition is scrubbed, never triggered.

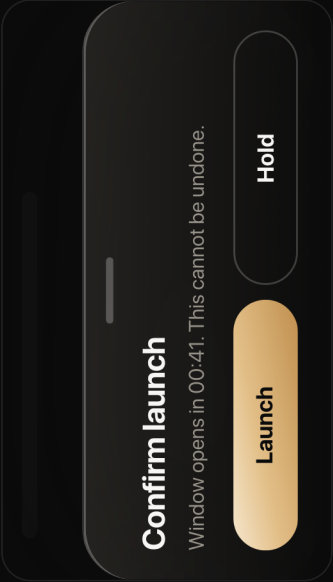
TAB BAR



Three to five destinations, never a verb among them. The bar is a floating capsule inset 10 pt from the screen edge — it hovers over content rather than terminating it.

Chrome is glass and floats; content is opaque and scrolls beneath it. The two never trade roles.

SHEET



Thick glass, radius- xl on the top corners only, grabber always present. The parent scales to 0.94 and dims — depth is signalled by the world receding, not by the sheet shouting.

ALERT



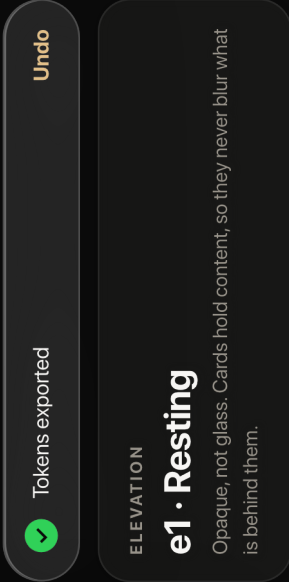
Two actions maximum. The title is a question and the body states the consequence in numbers. The safe action is on the left and never wears metal.

POPOVER & MENU



Scales from 0.92 anchored at its trigger's edge, so the eye never loses the origin. Destructive items sit last, below a divider, and are the only coloured rows.

TOAST & CARD

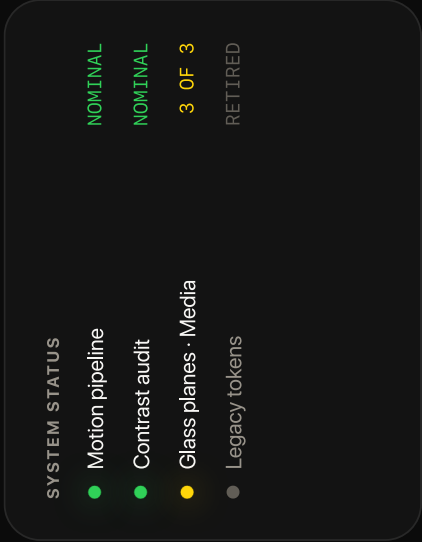


Toasts are capsules, live 4 s, carry one action, and stack to a maximum of two. A toast is never the only record of an event.

Telemetry

Numbers are the Frontier axis made visible. One number per view is allowed to be enormous.

Every figure is tabular, every unit is separated from its value, every precision claim is one the data can support.



TABLE

FLOW	P50	P99	Δ	STATE
Sheet present	4.12	6.91	-0.34 ✓	PASS
Hero morph	5.08	7.84	+0.61 !	WATCH
List appear x64	3.44	5.20	-1.08 ✓	PASS
Scrub · 120 Hz	2.90	4.02	-0.12 ✓	PASS
Glass x4 (invalid)	9.71	14.28	+5.95 ×	BLOCK

Header row repeats on every printed page. Numeric columns are right-aligned with tabular figures so the decimal points form a single vertical rule the eye can scan without reading.

PRECISION HONESTY

Show only digits the measurement supports. 6.91 ms claims ±0.005; 6.9142 ms claims an accuracy no sampler has, and a user who notices stops believing every other number on the screen.

UNITS ARE QUIETER

The unit is set at 0.28 × the value's size, in content-tertiary, separated by a hair space. The magnitude is the message, the unit is the footnote.

DELTA ALWAYS CARRIES A DIRECTION

A triangle plus a sign plus a colour — three redundant channels, because a change is the one figure a user acts on. "Better" is never assumed from "up".

CHARTS ARE INK-MINIMAL

No gridlines, no legend where a direct label fits, no axis the data does not need, no gradient that does not encode a value. One stroke, 2 pt, Aurum. Everything else is Titanium at 10%.

LIVE VALUES ROLL

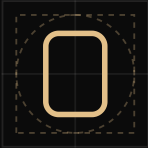
A changing digit translates vertically over 280 ms on Snap; the digits around it never move. Updates are throttled to 10 Hz however fast the source — a value nobody can read is noise.

Icons are type

SF Symbols only: An icon's weight matches the weight of the text beside it, always.

CONSTRUCTION GRID

FRAME	80 × 80 pt canvas, 64 pt live area, 8 pt optical margin on all sides.
STROKE	3 pt at 80, scaling linearly. Round caps, round joins, no mitred corner anywhere.
RADIUS	Inner corners 2 pt, outer 7 pt — continuous curvature, matching the component radii.
OPTICAL	Circular forms overshoot the live area by 2 pt; a circle drawn to the same box as a square looks smaller.



80 pt frame · 64 live

WEIGHT PAIRING

TEXT STEP	SYMBOL WEIGHT	POINT SIZE	SCALE
Outline / Caption	Semibold	11	Small
Body / Subhead	Regular	17	Medium
Headline / control	Semibold	17	Medium
Title 1–3	Bold	22–34	Large
Display / Colossus	Bold	48+	Large + hierarchical

A symbol set that shares its metrics with the typeface aligns on the baseline for free — and alignment for free is alignment that never regresses.

RENDERING MODES

Monochrome everywhere by default. **Hierarchical** — one hue at three opacities — for icons above 34 pt. **Palette** only for status glyphs paired with a Signal colour. **Multicolour** is never used: it imports a colour system this one has already rejected.

NEVER DRAW A NEW ONE

If the concept exists in SF Symbols, use it — a user has already learned it in nine other apps and recognition is free. Custom symbols are permitted only for brand-specific objects, and must be built on this grid with the same stroke, joins and optical rules.

NEVER AN ICON ALONE

Icon-only controls are limited to eight universally learned glyphs — close, back, share, search, add, more, play, settings. Everything else carries a visible label. An icon quiz is not minimalism; it is a tax the user pays on every visit.

FILLED MEANS SELECTED

Outline is the resting state; the filled variant is the selected state. This is the only meaning fill carries, and it is never inverted for visual variety.

SYMBOL ANIMATION

One effect per event, 320 ms, on Snap: **bounce** for confirmation, **replace** for a state change, **variable** for a level, **pulse** for an ongoing process. Nothing loops unless the underlying process does.

ALIGNMENT

Icons align to the text **baseline**, not to the line box, and are optically centred within their capsule rather than mathematically centred — a triangle centred by its bounding box always looks pushed left.

The felt layer

Haptics are the only channel that reaches the user without being looked at. They are the closest an interface gets to weight.

The transient fires on the **visual keyframe**, not on the touch. Bound within 60 ms, sight and touch fuse into one event; outside it, they read as two.

PATTERN	SHARP	INTENSITY	DURATION	FIRED ON	PAIRED SOUND
Light	0.4	0.35	12 ms	Capsule contact, tab, chip, drag lift	Silent
Rigid	0.9	0.55	10 ms	Switch detent, picker notch, snap-to-grid	Tick · 2.4 kHz · -34 dB · 18 ms
Soft	0.1	0.45	45 ms	Sheet arrival, toast, soft landing	Air · 180–400 Hz · -40 dB · 220 ms
Medium	0.6	0.70	20 ms	Drop landing, refresh trigger, threshold crossed	Seat · 900 Hz · -32 dB · 60 ms
Success	0.5 → 0.8	0.6 → 1.0	2 × 90 ms	Completion — at 70% of the checkmark draw	Rise · two tones, perfect fifth · -30 dB
Warning	0.7	0.8	2 × 60 ms	Limit approached, unsaved exit	Silent — visual and haptic only
Error	0.9	1.0	3 × 55 ms	Failure — on the first shake pass	Fall · two tones, minor third · -28 dB
Singular	0.2 → 1.0	0.3 → 1.0	640 ms	The one irreversible act in the product	Bloom · 40–1800 Hz sweep · -26 dB · 900 ms

BUDGET

Four haptics per minute of ordinary use. Beyond that the actuator becomes background noise and the user turns the whole channel off — losing the ones that mattered.

SILENCE IS THE DEFAULT

Only five events in the entire system make a sound, none above -26 dB, none longer than 900 ms. Every sound is optional and respects the silent switch without exception.

MATERIAL VOICE

Sounds are recorded from real objects — a titanium plate, a sapphire crystal, a machined detent — then filtered, never synthesised from a waveform. The ear knows the difference even when it cannot say why.

NEVER THE ONLY CHANNEL

No state is communicated by haptic or sound alone. Both are amplifiers of something already visible — and both are switched off for a meaningful share of users.

Say less, mean more

The system speaks like an engineer who respects you: precise, unhurried, never charming.

FOUR RULES

- 1 **Name the consequence, not the mechanism.** "Nine products reference this theme" beats "Are you sure?" every time.
- 2 **Use the number.** A specific figure is the cheapest possible proof of competence. Vagueness reads as either ignorance or evasion.
- 3 **Never apologise, never celebrate.** No "Oops", no "Great job!". The system did its work; the user did theirs. Neither needs a performance.
- 4 **One idea per string.** If a label needs a comma, it is two labels. If a sentence needs a semicolon, it is two sentences.

FORBIDDEN VOCABULARY

Oops

Simply

Whoops

Just

Awesome

Seamless

Something went wrong

Powerful

Revolutionary

Copy is the highest-leverage surface in any interface. A word costs nothing to change and everything to get wrong.

REWRITES

Oops! Something went wrong:

Upload stopped at 82%. Your connection dropped for 4 seconds.

→ names the state, the point of failure and the cause

Are you sure you want to delete?

Delete 41 frames. This cannot be undone.

→ the count is the warning; the button carries the verb

Please wait while we process your request...

Rendering 412 tokens · 00:09 remaining

→ waiting is tolerable; not knowing is not

No items to display

Nothing here yet. Your first theme takes about a minute.

→ an empty state is an invitation, never a report

Sentence case, always. Title Case In Buttons Reads As Advertising. Only proper nouns and the first word are capitalised — including in menus and tabs.

Numerals over words. "3 minutes", not "three minutes". Digits are scanned; words must be read.

Second person, active voice. "You have 4 drafts", not "There are 4 drafts". The user is the subject of every sentence in their own software.

The floor, not the ceiling

Nothing on this sheet is optional, and nothing on it may be traded for visual effect.

A system that becomes unusable when a setting is changed was never a system — it was a screenshot.

SETTING	WHAT THE SYSTEM DOES	VERIFICATION
Reduce Motion	All travel and scale become a 160 ms opacity cross-fade at the destination. Parallax, auto-play and the ambient glow loop stop. No state is lost.	Every flow completable with zero translate keyframes
Reduce Transparency	Every glass recipe collapses to an opaque token fill at equal luminance. The rim survives; the blur does not.	Zero backdrop-filter in the computed tree
Increase Contrast	Every label promotes one rung up the ramp; rim opacity doubles; hairlines become 1 pt at 24%; tinted fills become bordered.	All text ≥ 7.0:1, all control edges ≥ 4.5:1
Larger Text · to AX5	Type scales across a 310% range. Horizontal rows restack vertically past 235%. Capsules grow in height, never shrink their label.	No truncation, no clipped glyph, no target under 44 pt
Bold Text	Regular becomes Semibold, Semibold becomes Bold, symbol weights follow. Layout must absorb the extra width without reflowing.	Same line count at both weights
VoiceOver	Every control has a label, a role, a value and a trait. Decorative glass, glows and dividers are hidden from the tree entirely.	Full task completion with the screen off
Keyboard & switch	Focus order follows visual order. The 3 pt Aurum ring at 2 pt offset is never removed, never inset, never replaced by a colour change.	Every action reachable by Tab alone
Colour vision	Every Signal use carries a shape, a word or a position as a second channel. Nominal and Abort are never distinguished by hue alone.	Legible in greyscale and in all three dichromacies
Motor	No gesture is the only path to an action. No timing requirement under 5 s. Destructive actions sit ≥ 24 pt from confirming ones.	Every flow completable with single taps only

WHY THIS BELONGS HERE

Accessibility is not a compliance annex to a luxury system — it is the luxury. An object that works for a wider range of hands, eyes and conditions is simply a better-engineered object.

THE DARK-THEME TRAP

Obsidian canvases invite thin, low-contrast type because it looks elegant in a mockup. On a sunlit train it disappears. Every value on sheet 24 was measured, not judged by eye in a dark room.

ENFORCEMENT

Contrast, target size, focus order and Dynamic Type are asserted in CI. Screen-reader traversal of the twelve core flows is a manual gate before every release. A failure blocks the build, not the backlog.

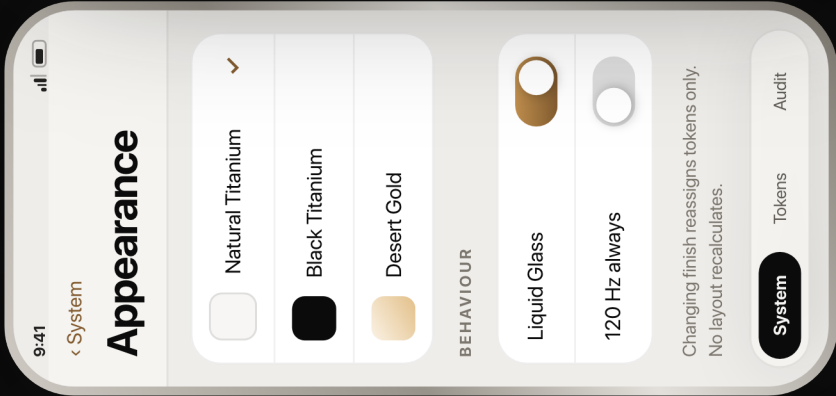
Three screens, one system

Nothing below uses a value, radius, curve or weight that is not specified in this document.



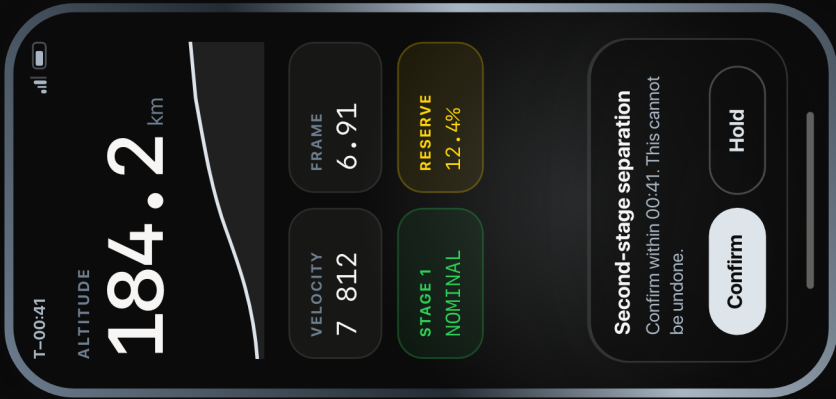
BLACK TITANIUM

Colossus is spent on nothing; the mark is the peak. One Metal capsule, one glow, e5 used once in the entire product.



NATURAL TITANIUM

Grouped fills on a Ti-05 canvas, Thin glass chrome top and bottom, Aurum 70 as the only accent — legible at 5.49:1.



VAPOR CHROME

Gold suspended. Cold light carries hierarchy; Obsidian glass holds the irreversible action, every figure is tabular mono.

Three finishes, three densities, one unmistakable family resemblance.

WHAT MAKES THEM SIBLINGS

Identical capsule heights, identical concentric radii, identical 4-pt rhythm, identical curve family. Only the token assignment changes — and that is the entire point of a token layer.

ONE PRIMARY EACH

Begin · the selected finish row · Confirm. Every screen answers "what is this for?" with exactly one filled capsule, and the eye finds it before the first word is read.

CHROME FLOATS, CONTENT SCROLLS

In all three, the bars are glass inset from the edge and the content passes beneath them. Nothing is docked, nothing is a frame.

WHERE THE BUDGET WENT

Three glass planes total per screen. One glow, on one screen, once. One Colossus-scale figure per view. Everything else is Titanium doing quiet work.

Cover the labels and the three screens are still recognisably one product. That is the only test of a design system that matters.

The primitive layer

Copy-ready custom properties. Semantic roles map onto these; product code touches only the roles.

COLOUR · TITANIUM

- ti-00: #F7F6F4;
- ti-05: #EFEDEA;
- ti-10: #E3E0DB;
- ti-20: #CFBCA4;
- ti-30: #B4AFA7;
- ti-40: #98938A;
- ti-50: #7C776E;
- ti-60: #615D56;
- ti-70: #4A4741;
- ti-80: #35322E;
- ti-90: #232120;
- ti-95: #171615;
- ti-100: #0B0B0B;

COLOUR · AURUM

- au-05: #FBF3E6;
- au-10: #F6E7CE;
- au-20: #EED6AE;
- au-30: #E3C089;
- au-40: #D4A96A;
- au-50: #C08F4E;
- au-60: #A5763D;
- au-70: #835C30;
- au-80: #5E4223;
- au-90: #3A2916;

COLOUR · VAPOR & SIGNAL

- az-10: #DDE4EA;
- az-30: #A9B6C4;
- az-50: #6E7F90;
- az-70: #44505C;
- az-90: #232B33;
- sig-nominal: #30D158;
- sig-caution: #FFD60A;
- sig-abort: #FF453A;
- sig-link: #0A84FF;

TYPE

- font-display: "SF Pro Display";
- font-text: "SF Pro Text";
- font-mono: "SF Mono";
- size-colossus: 96px/88px;
- size-display-1: 64px/64px;
- size-display-2: 48px/52px;
- size-title-1: 34px/40px;
- size-title-2: 28px/34px;
- size-title-3: 22px/28px;
- size-headline: 17px/22px;
- size-body: 17px/25px;
- size-callout: 16px/21px;
- size-subhead: 15px/20px;
- size-footnote: 13px/18px;
- size-caption: 12px/16px;
- size-micro: 11px/13px;
- size-overline: 10px/12px;

SPACE & RADIUS

- space-1_16: 4 8 12 16 20 24 32 48 64;
- radius-sm: 12px;
- radius-md: 16px;
- radius-lg: 20px;
- radius-xl: 28px;
- radius-capsule: 999px;
- corner-shape: square;
- target-min: 44px;

MOTION

- ease-standard: cubic-bezier(.32,.72,0,1);
- ease-enter: cubic-bezier(.16,1,.3,1);
- ease-exit: cubic-bezier(.7,0,.84,0);
- ease-glide: cubic-bezier(.4,0,.2,1);
- ease-snap: cubic-bezier(.34,1.26,.64,1);
- dur-micro: 90ms;
- dur-small: 160ms;
- dur-medium: 240ms;
- dur-large: 400ms;
- dur-enter: 520ms;
- dur-morph: 620ms;
- dur-ambient: 4000ms;
- stagger: 24ms;
- frame-budget: 8.33ms;
- spring-tap: 1 500 32;
- spring-snap: 1 420 26;
- spring-sheet: 1 240 28;
- spring-morph: 1 180 22;
- spring-scrub: 1 300 34.6;
- spring-inertia: 1 220 26;

MATERIAL & ELEVATION

- glass-thin: blur(20px) saturate(170%);
- glass-regular: blur(28px) saturate(180%);
- glass-thick: blur(40px) saturate(150%);
- e1: 0 4px 2px / 0 0 1px;
- e2: 0 4px 10px / 0 1px 2px;
- e3: 0 10px 24px / 0 2px 5px;
- e4: 0 18px 44px / 0 4px 10px;
- e5: 0 28px 70px + glow 42px;

aurum.tokens.css · 27.0
412 declarations · 0 aliases unresolved

Nothing Every value was made defined

A design system is not a library of components. It is a written agreement about what will never change
— so that everything built on top of it can change freely, and still be recognised a hundred years from
now as the work of one hand.

EDITION

27.0 · MMXXVI
46 sheets · 412 tokens

SET IN

SF Pro Display · Text
SF Mono · tabular

PALETTE

Titanium · Aurum
Vapor · Signal

GOVERNANCE

Additive only. A token
is retired, never edited.

CONTRACT

8.33 ms · 4.5:1
44 pt · one gold

